Geometry Coordinate Geometry & Transformations Cheat Sheet

Unit Focus: formulas, graphs, and transformations

1 COORDINATE PLANE

An ordered pair (x, y) locates a point.

x: horizontal position; y: vertical position.

Origin: (0, 0)

Quadrants:

I: (+,+) II: (-,+)

III: (-,-) IV: (+,-)

2 DISTANCE FORMULA

Distance between two points:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: P(1, 2), Q(5, 5)

$$d = \sqrt{4^2 + 3^2} = 5$$

3 MIDPOINT FORMULA

Average the x- and y-coordinates.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example: A(2, 1) and B(8, 5)

$$M = \left(\frac{2+8}{2}, \frac{1+5}{2}\right) = (5, 3)$$

4 SLOPE

Slope = rise divided by run.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Example: (1, 2) and (4, 6)

$$m = \frac{6-2}{4-1} = \frac{4}{3}$$

5 PARALLEL & PERP. LINES

Parallel nonvertical lines:

Same slope ($m_1 = m_2$).

Perpendicular nonvertical lines:

Slopes are negative reciprocals.

$$m_1 m_2 = -1$$

6 TRANSFORMATIONS

Translation: $(x, y) \to (x + a, y + b)$

Reflection in x-axis: $(x, y) \to (x, -y)$

Reflection in y-axis: $(x, y) \to (-x, y)$

180° rotation: $(x, y) \to (-x, -y)$

Dilation, scale k: $(x, y) \to (kx, ky)$

7 GRAPHS & QUICK REMINDERS



- ✓ **Distance is never negative.**
- ✓ **Mark and label points carefully.**
- ✓ **Match corresponding coordinates.**
- ✓ **For slope, subtract in the same order.**

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More Practice: reflections, rotations, dilations, and sequences

1 TRANSLATION

A **translation** slides a figure without changing its size or shape.

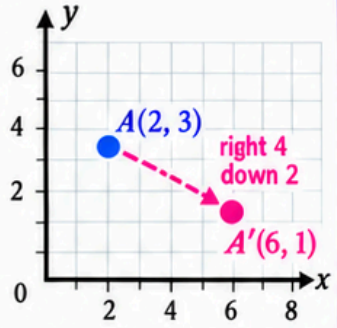
$$(x, y) \to (x + a, y + b)$$

Example:

Translate point $A(2, 3)$ right 4 and down 2.

$$(x, y) \to (x + 4, y - 2)$$

$$A(2, 3) \to A'(6, 1)$$

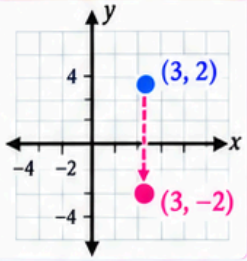


2 REFLECTIONS

Reflections create a mirror image across a line.

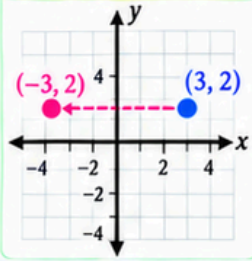
Over the x-axis

$$(x, y) \to (x, -y)$$



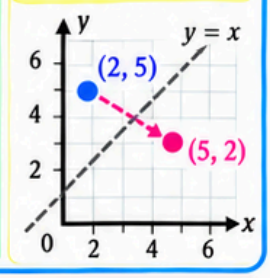
Over the y-axis

$$(x, y) \to (-x, y)$$



Over the line $y = x$

$$(x, y) \to (y, x)$$

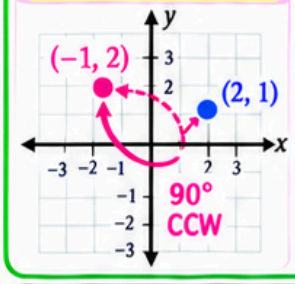


3 ROTATIONS ABOUT THE ORIGIN

A **rotation** turns a figure around the origin (0, 0).

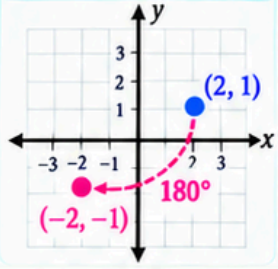
90° Counterclockwise

$$(x, y) \to (-y, x)$$



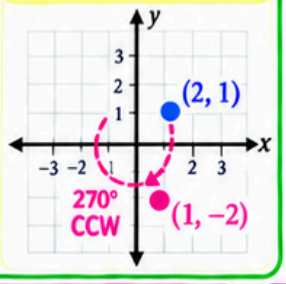
180° Rotation

$$(x, y) \to (-x, -y)$$



270° Counterclockwise

$$(x, y) \to (y, -x)$$



4 DILATIONS

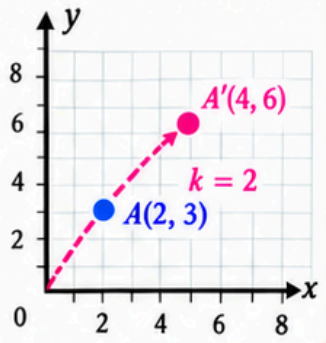
A **dilation** from the origin multiplies the coordinates by a scale factor k.

$$(x, y) \to (kx, ky)$$

Example: Dilate point $A(2, 3)$ by a scale factor of $k = 2$.

$$(x, y) \to (2x, 2y)$$

$$A(2, 3) \to A'(4, 6)$$



5 CONGRUENCE VS. SIMILARITY TRANSFORMATIONS

Congruence Transformations

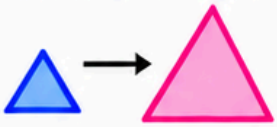
Translations, reflections, and rotations preserve size and shape. The image is congruent to the original figure.



Same size and shape (congruent)

Similarity Transformations

A dilation preserves shape but changes size unless $k = 1$. If $k = 1$, the figure is congruent.



Same shape, different size (similar)

6 SEQUENCE OF TRANSFORMATIONS

You can apply more than one transformation in a row.

Example: Translate then reflect.

Start with $A(1, 2)$.

Step 1: Translate right 3.

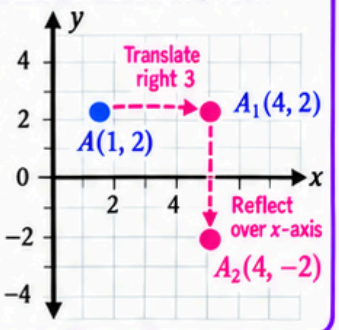
$$(x, y) \to (x + 3, y)$$

$$A(1, 2) \to A_1(4, 2)$$

Step 2: Reflect over the x-axis.

$$(x, y) \to (x, -y)$$

$$A_1(4, 2) \to A_2(4, -2)$$

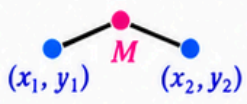


7 COORDINATE GEOMETRY REVIEW

Midpoint Formula

The midpoint between (x_1, y_1) and (x_2, y_2) is

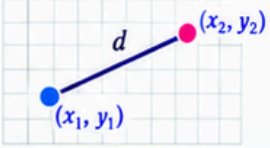
$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$



Distance Formula

The distance between (x_1, y_1) and (x_2, y_2) is

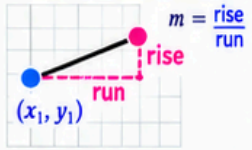
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Slope Formula

The slope of a line through (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Ordered pairs are written (x, y) (x comes first, then y).
- Be careful with signs when adding, subtracting, or reflecting.
- Use the correct rules for 90°, 180°, and 270° rotations.
- A reflection over $y = x$ swaps the coordinates.
- Dilations multiply both coordinates by k.

COMMON MISTAKES:

- Reversing the order of an ordered pair (writing y, x instead of (x, y)).
- Forgetting to change both coordinates in a transformation.
- Confusing the 90° and 270° rotation rules.
- Thinking a reflection over $y = x$ keeps the coordinates the same (instead of swapping).
- Forgetting that dilations multiply both x and y by k.